Incomplete Swallowing and Retracted Tongue Maneuvers for Electromyographic Signal Normalization of the Extrinsic Muscles of the Larynx

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Summary: Purpose. To investigate which muscular maneuvers provide larger electric activity (EA) of the suprahyoid (SH) and infrahyoid (IH) muscles to be used as surface electromyography (SEMG) signal normalization reference.

Methods. The electrical potentials of the SH and IH muscles of 12 subjects were evaluated using six muscular maneuvers, involving the position of the tongue and effort. It was selected as maximum voluntary sustained activity maneuver, the one having the minor coefficient of variation and the smallest value for each muscle group. The EA signal was converted using the root mean square in microvolts. It was considered then the maximum signal of each maneuver as the difference between the mean of three measures and the resting potential.

Results. The maneuvers that provided higher mean potentials with minor coefficient of variation and smallest *P* value were incomplete swallowing (IS) with effort (mean potential equal to 56.73 ± 8.68 with coefficient of variation of 15.30%) in SH group, and tongue retracted with mouth open (TROM, mean potential equal to 46.57 ± 7.83 with coefficient of variation of 16.81%) in IH group.

Conclusion. The IS with effort and TROM maneuvers should be considered for signal normalization in these muscles, respectively, and may provide conditions for using the SEMG in voice clinic.

Significance. The use of normalization standards in researches of SH and IH muscles in the voice area will allow comparisons among future works.

Key Words: Electromyography–Voice–Phonation–Laryngeal muscles.

INTRODUCTION

Surface electromyography (SEMG) is a noninvasive procedure used to help the diagnosis and follow-up of the electrical activities (EAs) variations of muscular groups. This method has been amply used in neurology, orthopedics, physiotherapy, and physical education for evaluation of the muscles or muscular groups of interest in these areas, observing the recommendations of the SEMG for the noninvasive assessment of muscles (SENIAM). The purpose of these recommendations is to integrate basic research in the area and to enable the exchange of data and experience.[1]

In speech and language pathology, the SEMG has been much more used in researches about chewing, swallowing, and alterations of language, that is, stuttering, for investigation of various muscular groups.[2–4] Suprahyoid (SH) and infrahyoid (IH) extrinsic muscles, as in normal phonation as in dysphonia cases, may be investigated using this procedure, but researches are few and hard to compare because the technique, size sample, and muscle groups used are different among papers.[2,3,5–8]

In EA evaluation by SEMG, a reference pattern of muscular contraction is used, entitled normalization signal, used to reduce inter- and intrasubject variability. There are several possibilities of normalization signal, by the maximum peak of the EA as well as by the maximum voluntary isometric contraction (MVIC), referential demanding the measurement of determined applied load, such as resistance of the studied muscle.[7–9]

Electromyography (EMG) normalization signal by MVIC has been vastly discussed and referred as the most adequate one, taking the value of referred sustained contraction as the EA maximum percentage of the muscle or muscular group to be studied on each subject. It allows the comparison between the value found in the MVIC and the percentages acquired with performed tasks.[7]

MVIC is an isometric task, which serves as reference for the normalization of EA of most muscles. It should be performed with a known static load, in which the muscle must not vary in length.[10] However, in the larynx, even in situations of isometric effort, the MVIC should not be considered as normalization because the plasticity of the phonatory system does not allow load application. Nevertheless, the importance of the signal normalization by maximum contraction is admitted to evaluate dysphonic subjects whose phonation may request the excessive contraction of the SH and IH extrinsic muscles, in a compensatory manner to an intrinsic muscles dysfunction.[11]

An alternative to this problem may be to consider the maximum contraction of the SH and IH extrinsic muscles as the functionally maximum voluntary sustained activity (MVSA). The MVSA can be defined as the muscular contraction generated

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when the subject recruits, voluntarily, an expressive number of the SH and IH fiber muscles, during a determined period of time in isometric tasks.[9] This is an adaptation of the concept of maximum voluntary resisted activity proposed by Moraes et al.[9]

The purpose of this study was to investigate which muscular movements provided larger SH and IH muscles EA that could be used as MVSA reference for SEMG signal normalization.

MATERIALS AND METHODS

Twelve subjects of both sexes, aged between 18 and 45 years, who did not have dysphonia, hearing loss, and reported history of cervical musculoskeletal disorders were evaluated. To collect the electrical potentials of the SH and IH muscular groups, evaluated in microvolts, MIOTOOL 200 EMG (Rio Grande do Sul, Brazil) was used, providing the possibility of the section of eight independent channels, using a 1000 μV gain. Three bipolar SDS500 sensors connected through claws, a reference cable, calibrator, the *Miograph* 2.0 software (Rio Grande do Sul, Brazil), and an USB communication cable were used, all of the MIOTEC brand (Rio Grande do Sul, Brazil), with infant disposable surface electrodes, MEDITRACE brand (Kendall, Graphic Controls, Canada). EMG signals were band-pass filtered at 10–500 Hz. Signals were sampled at 1000 Hz.

During EMG evaluation and register, subjects were seated in a chair, with erect torso, eyes open, feet resting on the floor, arms resting on their legs, backs to the equipment, to prevent any attempt of monitoring or visual feedback. There was no need of subjects' trichotomy. The area was cleaned with gauze soaked in ethyl alcohol at 70%, whose abrasion is larger than cotton. The electrodes were allocated: one channel with two electrodes in the submandibular region to obtain signals of mylohyoid and digastric muscles; and two channels disposed bilaterally to the larynx, at 1 cm from thyroid incisure,[12] and the interelectrodes distance was equal to 2 cm as recommended by SENIAM[1] (Figure 1). An EMG was connected to the notebook, LG brand (São Paulo, Brazil). To avoid external electrical noise interference, an electrode of reference ("ground") was placed on right arm ulnar styloid process, far from the evaluated muscles.[13]

Six muscular maneuvers were tested: incomplete swallowing (IS) with effort; tip of the tongue stuck to the palate, with effort and open mouth (TTPOM) and closed (TTPCM); and tongue retracted with open mouth (TROM) and closed (TRCM), and pushing the wall. The maneuvers were selected according to the classification of approach of vocal training proposed by Behlau et al,[13] adapting some of the techniques in which noise emissions have been excluded because the purpose was not to evaluate EA of larynx intrinsic muscles. The IS with effort was adapted from the incomplete sonorous swallowing, which physiologically produces the laryngeal closing and the forced adduction of the vocal folds. The sonorous plosion was excluded, including for IS the sustained contraction with effort to test its response in SH and IH groups. The TTPOM and TTPCM were selected because they provide big pressure on the muscles of the mouth floor.[10] The TR (OM and CM) is an adaptation of the tongue shift posteriorization-type technique, which provides, among other adjustments, the laryngeal lowering and the activation of the IH group. The pushing wall is derived from the effort technique, which provides laryngeal closing and stability.

First, the participants were asked to close their eyes and relax during 1 minute. The electrical signal determined during this minute was the resting potential (R), considered as a baseline reference for all maneuvers.

For each maneuver, three measurements were performed with the maximum of sustained contraction lasting 5–7 seconds, under verbal command "Attention! Contract! Maintain! Maintain!" following 10 seconds of rest between each measurement. Between one type of maneuver and the next, the subject was asked to relax as much as possible, by the verbal command "Relax! Relax! Let your body go and just breathe…" to avoid measurement bias owing to the sustained contraction.

All maneuvers were tested separately for SH and IH muscular groups because these groups, even both having traction the larynx, are antagonists. SH are responsible for the elevation of the hyoid bone and, consequently, of the larynx; and IH, responsible for the lowering of this structure.

Because a proper SEMG temporal window is close to 1 second, the entirety of each maneuver was used for each EA measure, resulting in a variable temporal window. To locate periods in which

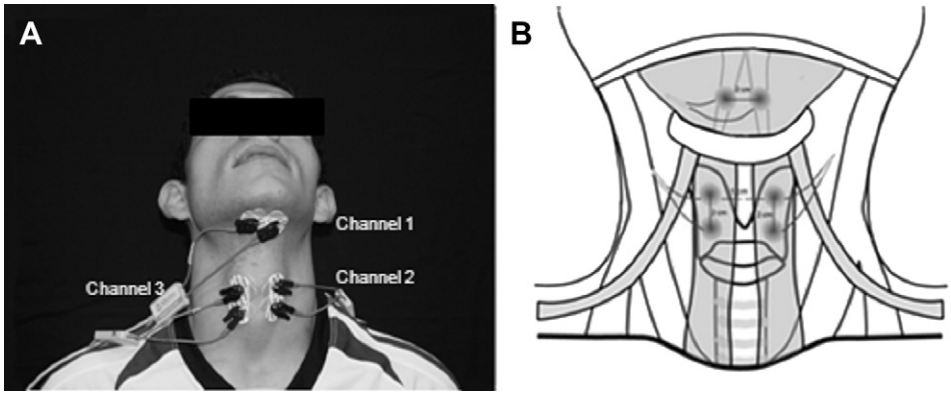

FIGURE 1. Allocation sites of the electrodes. Note: The electrodes were allocated in the submandibular region (floor of the mouth) and two channels disposed bilaterally to the larynx, at 1 cm from thyroid incisure to infrahyoid muscles.

muscles were contracted, a visual inspection of the electrical signal associated to the temporal registration of the beginning and the end of contraction was used. To reduce the interference of other sources of electrical noise, line filters were used.

The EA signal was measured during 3-seconds intervals, annulling the first and the last second because these seconds are characterized by major instability. The EA was converted in microvolts, using the root mean square (RMS). To determine the EA restricted to each maneuver, we considered the difference between the mean of three measures and the resting potential. The IH muscles group maneuvers potentials were the mean between right and left, in each individual. For characterization of each maneuver, one has considered mean, mean standard error, and the coefficient of variation of the 12 subjects, expressed in microvolts. The entire experimented lasted 25–30 minutes.

In both groups, the sign test was used, by two-by-two pairing of the maneuvers in level of significance of 5% because the distribution was not normal. It was elected as MVSA, aiming for the normalization of the signal, the maneuver that, in the muscular group, presented minor coefficient of variation and larger statistic significance.

The study was approved by the Ethics Committee in Research of the Federal University of Pernambuco, under the protocol number 029/2009.

RESULTS

In Table 1, the results point to: IS was the maneuver that favored the increase of EA in the SH group, having differed significantly from the TROM ($P = 0.006$) and TRCM ($P = 0.001$) maneuvers. The movement of larynx in this maneuver can be seen in Figure 2.

In Table 2, the maneuvers electrical potentials means for right and left IH muscular group are expressed, among which the TROM provided larger EA with smaller coefficient of variation, differing significantly from TTPOM ($P = 0.034$) and TTPCM ($P = 0.015$) maneuvers. The movement of larynx in this maneuver can be seen in Figure 3.

In Figure 4, it is possible to observe that the maneuvers IS, TTPOM, and TTPCM requested predominately the SH muscular group, differing from the TROM, TRCM maneuvers, and pushing, in which the IH group was one that had the highest EAs.

Figure 5 shows the electrical signals in RMS according to maneuvers for SH and IH muscle groups. Note that the signal configurations captured with SEMG are greatly variable owing to the constant automatic alterations of the amplitude scale during signal acquisition. It means that major peaks not necessarily correspond to major values of SEMG.

DISCUSSION

Because of its uniqueness, the research here presented proposes IS and TROM to be used as clinically feasible maneuvers, for SEMG EA signal normalization. The physiology of these techniques is visually and tactually perceptible because both have antagonistic traction on the larynx (Figures 2 and 3). The IS maneuver lifts the hyoid bone and the TROM lowers it. With the results of this research, it is clear how these muscles become activated when the tongue is contracted, lifted, and sustained, as in IS, TTPOM, and TTPCM, as well as in the retraction of the tongue, as in the TROM and TRCM. The studies with the swallowing function use various types of lingual pressure to normalization; however, there is no reference about which of these produces greater activity.[12,14]

Obviously, the SEMG is not specific to identify the activation of specific muscles. Nevertheless, it is also obvious that the maneuvers here proposed elicit SH and IH groups to be performed; so, based on our data presented in Tables 1and 2, it seems proper to admit that these maneuvers are useful to normalize SEMG signal.

As to the lingual retraction, opening the mouth variation was used as a strategy to facilitate the visualization of the subject

TABLE 1.
Comparison of the Maneuver Potentials for the SH Muscle Group

Maneuvers of Reference	Mean (SE, μV)	IS, N (*P* Value)	TTPOM, N (*P* Value)	TTPCM, N (*P* Value)	TROM, N (*P* Value)	TRCM, N (*P* Value)	Pushing, N (*P* Value)
IS	56.7 (8.7)	—	6 (1.000)	8 (0.388)	**11 (0.006)**	**12 (<0.001)**	**11 (0.006)**
CV(%)	15.3	—	—	—	—	—	—
TTPOM	45.6 (7.0)	6 (1.000)	—	5 (0.774)	8 (0.388)	8 (0.388)	**11 (0.006)**
CV(%)	15.5	—	—	—	—	—	—
TTPCM	52.0 (8.1)	8 (0.388)	5 (0.774)	—	9 (0.146)	7 (0.774)	**10 (0.012)**
CV(%)	15.6	—	—	—	—		
TROM	36.2 (8.5)	**11 (0.006)**	8 (0.388)	9 (0.146)	—	6 (1.000)	9 (0.146)
CV(%)	23.6	—	—	—	—	—	—
TRCM	35.8 (6.0)	**12 (<0.001)**	8 (0.388)	7 (0.774)	6 (1.000)	—	9 (0.146)
CV(%)	16.8	—	—	—	—	—	—
Pushing	18.0 (4.0)	**11 (0.006)**	**11 (0.006)**	**10 (0.012)**	9 (0.146)	9 (0.146)	—
CV(%)	22.3	—	—	—	—	—	—

Abbreviations: SH, suprahyoid; IH, infrahyoid; IS, incomplete swallowing with effort; TTPOM, tip of tongue against palate with open mouth; TTPCM, tip of the tongue against palate with closed mouth; TROM, tongue retracted with open mouth; TRCM, tongue retracted with closed mouth; CV, coefficient of variation. The numbers in boldface indicate significant *P* value.

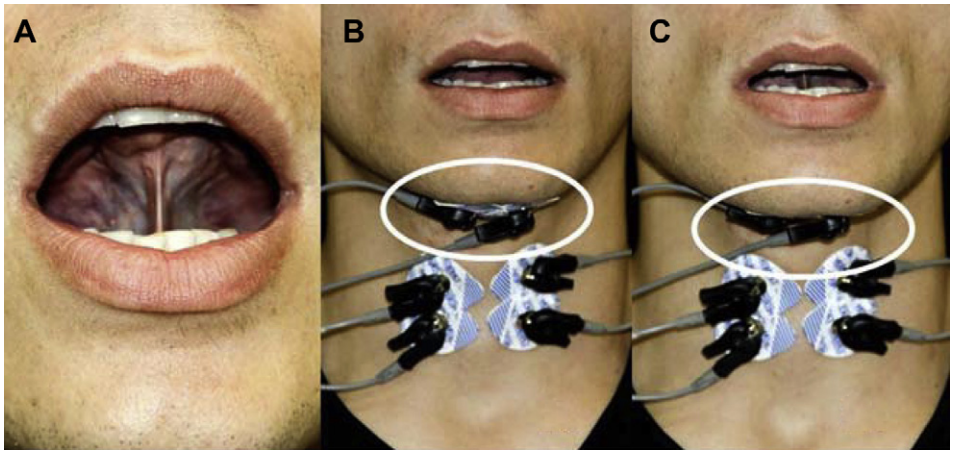

FIGURE 2. Difference of larynx position during incomplete swallowing with effort maneuver. Note: **A**. Incomplete swallowing with effort maneuver to normalize the electromyographic signal of suprahyoid muscles. There is a difference between the electrodes positions in (**B** and **C**), while the maneuver is performed, indicating larynx elevation.

during the completion of the maneuver, which finally sets up itself as being the one that caused higher activity. Therefore, the pressure of the tongue during the IS to normalize the signal in the evaluation of SH EA and the TROM for the IH group provides high EA in each group, respectively. Consequently, they can be used as reference for normalization of the EMG signal to be considered analogous to the MVIC, procedure here entitled as MVSA.

The present study allows some considerations about the use of the SEMG and the normalization signal and may confer greater objectivity to the evaluations of SH and IF muscles EA for studies of phonation and, perhaps, for other functions as swallow.

The use of SEMG requires the examiner's knowledge of the instrumentation as to the acquisition and signal analysis. Thus, the choice and the conditions of site to allocate the electrodes and the task being executed are determinant criteria in the quality of the examination. At the SENIAM recommendation to allocation electrodes, the shoulder and neck muscles are included, but there is no reference to SH and IH muscles. Therefore, we chose to use the location based on other studies.[15]

From anatomical standpoint, these muscles are covered by the quadrilateral subcutaneous muscle and by the platysma, more superficially. It involves a great part of this former area with its origin in the fascia of the chest muscle and enters in the inferior margin of the jaw, allowing tension to the skin of the neck; and dislocates the mouth angles down and helps the lowering of the jaws. Still, on the lateral region of the neck are mainly the sternocleidomastoid (SCM) and the scalenes that also have been object of investigation in the study of voice disorders.[5,7]

This muscular architecture of head and neck can limit the trustworthiness of the investigation of SH and IH muscles because, in the captured SEMG signal, cross talk can occur, which would be the interference by a signal of the adjacent muscles.[10]

As SEMG evaluates a pool of motor units and the human body works in a synergic manner, some measures are important as, for instance, the distance between the electrodes and the

TABLE 2.
Comparison of the Maneuvers Potentials for IH Muscle Group

Maneuvers of Reference	Mean (SE, μV) Right	Mean (SE, μV) Left	General Mean (SE, μV)	IS, N (*P* Value)	TTPOM, N (*P* Value)	TTPCM, N (*P* Value)	TROM, N (*P* Value)	TRCM, N (*P* Value)	Pushing, N (*P* Value)
IS	35,6 (6.0)	35.6 (6.0)	35.6 (5.0)	—	10 (0.158)	9 (0.239)	6 (0.272)	6 (0.530)	7 (0.638)
CV(%)	16.8	16.8	14.2	—	—	—	—	—	—
TTPOM	29.9 (11.8)	28.8 (12.4)	29.4 (12.0)	10 (0.16)	—	8 (0.158)	**11 (0.034)**	**11 (0.028)**	6 (0.433)
CV(%)	39.5	43.0	40.6	—	—	—	—	—	—
TTPCM	31.0 (9.5)	31.0 (10.0)	31.0 (9.6)	9 (0.239)	8 (0.158)	—	**11 (0.015)**	**10 (0.005)**	7 (0.695)
CV(%)	30.6	32.2	30.9	—	—	—	—	—	—
TROM	47.4 (8.6)	45.8 (7.2)	46.6 (7.8)	6 (0.272)	**11 (0.034)**	**11 (0.015)**	—	8 (0.347)	8 (0.182)
CV(%)	18.4	15.7	16.8	—	—	—	—	—	—
TRCM	42.2 (9.7)	44.4 (8.8)	43.3 (9.2)	6 (0.530)	**11 (0.028)**	**10 (0.005)**	8 (0.347)	—	7 (0.480)
CV(%)	23.0	19.8	21.2	—	—	—	—	—	—
Pushing	30.8 (7.0)	32.7 (8.0)	31.8 (6.8)	7 (0.638)	6 (0.433)	7 (0.695)	8 (0.182)	7 (0.480)	—
CV(%)	22.7	24.5	21.5	—	—	—	—	—	—

Abbreviations: SH, suprahyoid; IH, infrahyoid; IS, incomplete swallowing with effort; TTPOM, tip of tongue against palate with open mouth; TTPCM, tip of the tongue against palate with closed mouth; TROM, tongue retracted with open mouth; TRCM, tongue retracted with closed mouth; CV, coefficient of variation.

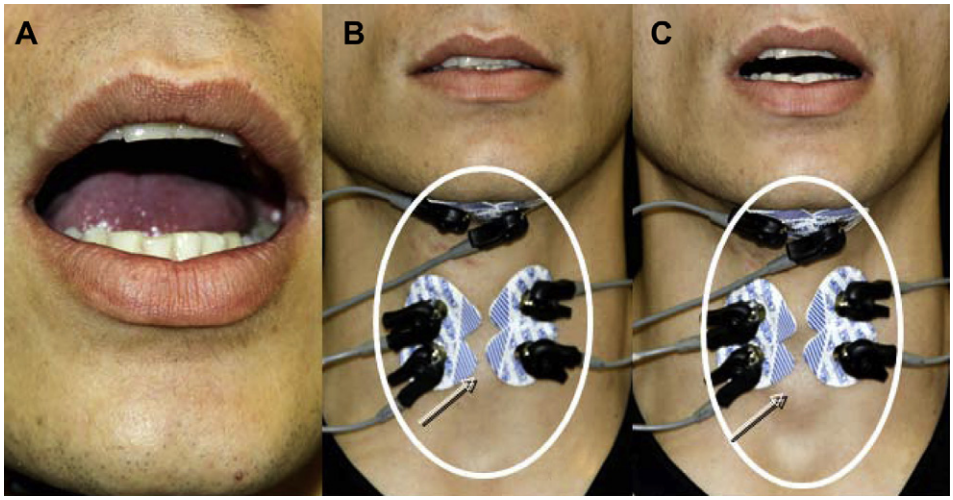

FIGURE 3. Difference of larynx position during a tongue retracted maneuver with open mouth. Note: **A**. Tongue retracted maneuver with open mouth to normalize the electromyographic signal of infrahyoid muscles. There is a difference at the electrodes position in relation to cricoid between (**B** and **C**), while performing the maneuver, indicating the debasing of larynx framework, without evident increased contraction of sternocleidomastoid muscle.

direction of putting them in line in the muscle fiber. The interelectrodes distance, from pole to pole (silver area) recommended by SENIAM, is up to 2 cm.[1]

To SH muscles, the allocations of electrodes proposed are in the direction of the fibers of the anterior venter of the digastric muscle, although we consider the muscular group as a whole. The studies in the orofacial motricity area have used, without reservations, this location to capture the signal during swallowing.[12,14–16]

For IH muscles, some questions must be considered. These are straight, thin muscles, extending from the hyoid bone to the sternum. They may suffer SCM overlay, depending on the anatomy of each person. In this way, to reduce cross talk and allow greater reliability of signal normalization, the present study was important because it shows that the action of the TROM causes high EA of the IH, under which the larynx lowers, when these are contracted.

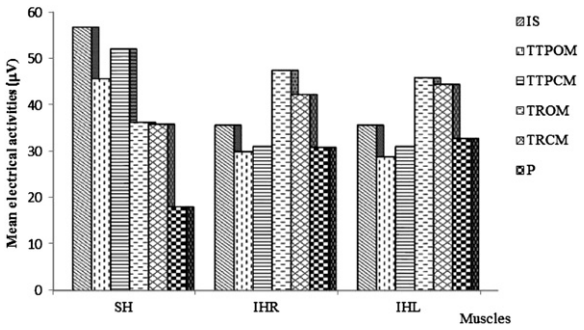

FIGURE 4. Mean electrical activity relative to muscle group during different maneuvers. SH, suprahyoid; IHR, right infrahyoid; IHL, left infrahyoid; IS, incomplete swallowing with effort; TTPOM, tip of tongue against palate with open mouth; TTPCM, tip of the tongue against palate with closed mouth; TROM, tongue retracted with open mouth; TRCM, tongue retracted with closed mouth; P, pushing the wall.

Although the study of Stepp et al[7] and Netto and Burnett[17] proposed the normalization of neck muscles by means of the MVIC with manual counterresistance, inclusively to investigate the EA in the hyperfunctional phonation,[7] this research allowed us to understand the maneuvers currently proposed can be more trustworthy in the evaluation of the muscles of interest. Moreover, the manual counterresistance can introduce variation capable to confer more bias to the tests because it provokes more tension, not only on the SCM but also on the nuchal muscles. This manual counterresistance, *per se*, recruiting a great number of muscles, can increase cross talk.

It is important to point out that in cases of more bulkier necks, the maneuver might not be efficacious, and, therefore, plausible to propose a more robust study, considering these anatomical differences.

The consensus in studies with SEMG is that the normalization decreases the variability among the subjects, as the functionality of a system differs among people and, by analogy, the same occurs in phonation. Thus, the subject is his/her own reference for the evaluation of the EA difference of SH and IH muscles during the phonation. Furthermore, considering the resting activity of each subject as the null value of reference, the number obtained during the maneuver represents the increased muscle function, expressed on rational scale. Thus, the possibility of increased resting activity or even hypertrophy of these muscles was nullified.

The possibility of using the normalization of the EMG signal of SH and IH muscles with MVSA will allow comparisons among future studies. Moreover, normalization by MVSA, and not by the peak contraction, can guarantee that, indeed, maximum muscle activity can be captured. Another important consideration is that the normalization by the peak may not behave in a periodic way, which is minimized in MVSA.

The capture of rest signal, for comparison purposes, is considered as reference of basal activity in the SEMG evaluation protocols. We consider that the rest should be done after

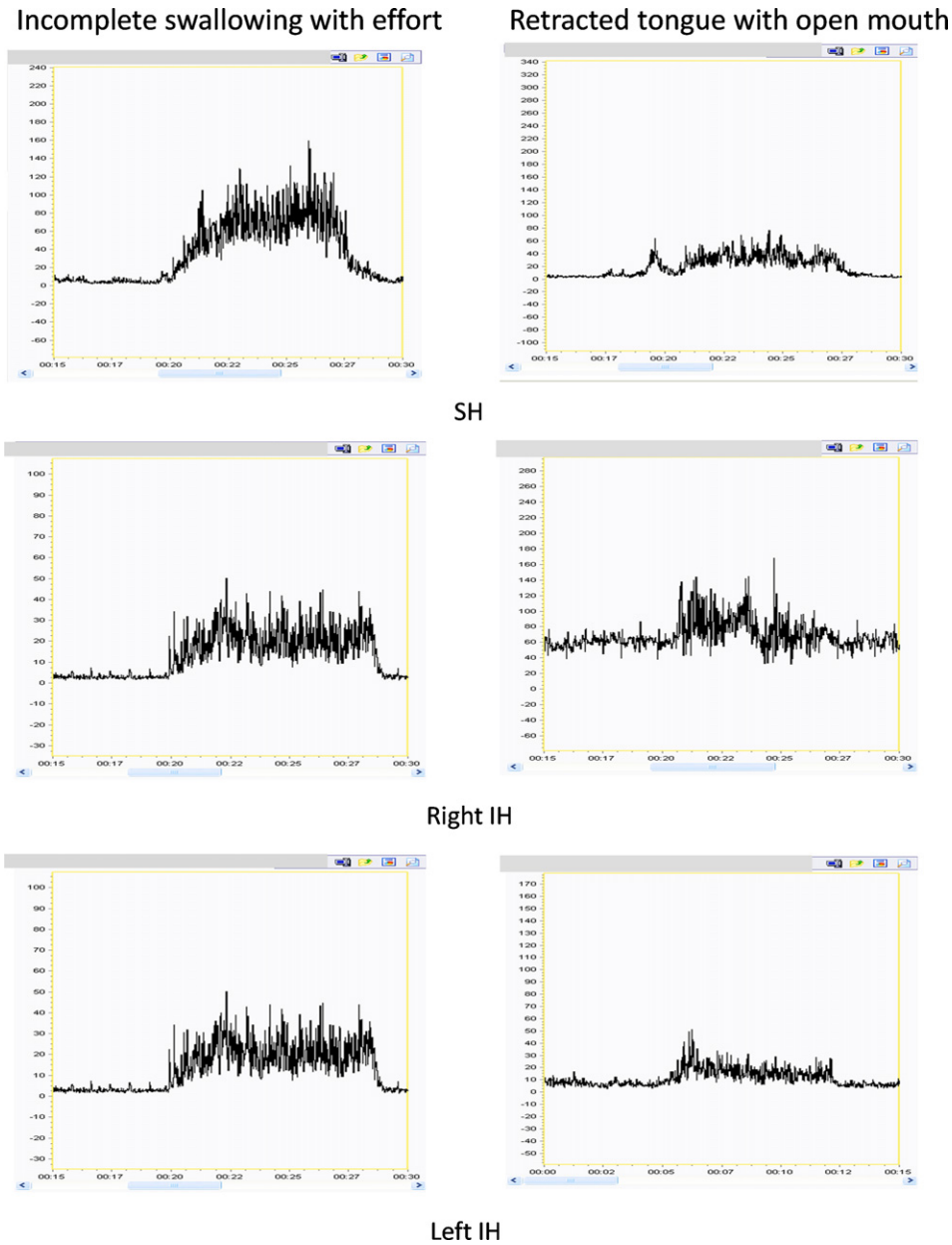

FIGURE 5. Electrical signals in RMS according to maneuvers for SH and IH muscle groups. SH, suprahyoid; IH, infrahyoid.

MVSA to guarantee that the subject is not in a situation of tension.[18]

Although we recognized that SEMG has limitations and demands technical care in the selection of electrodes location because it might not represent the EA for one unique motor unit, but rather signals of neighboring muscles, this is a procedure that will provide quantitative data in vocal assessment. Furthermore, from future works, which start using such techniques or maneuvers as normalization standard, comparisons among studies will be possible, qualifying the clinical application of SEMG in voice area.

CONCLUSION

The IS and TROM maneuvers should be considered for signal normalization of SH and IH muscles, respectively, by the MVSA because they show higher EA. In this way, adapting the principals recommended for the use of the SEMG for the voice clinic is feasible, if the correlation with the clinical history of the subjects and the pathophysiology of vocal disorders are considered and accompanied on therapeutic intervention.

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